

Applied A.I. Solutions

Data Visualization Techniques

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DATA VISUALIZATION TECHNIQUES

Business Analytics

Business Analytics

- Business Analytics help organizations uncover hidden patterns and trends and see the story that lives within their data to get relevant insights for planning, forecasting, informed decision-making and performance management.

Types of Analytics – Questions to answer

- what happened and why is this happening?
- how many and how often and what if these trends continue?
- where the problem is and what actions are needed?
- what will happen next? (predict)
- what is the best that can happen? (optimize, simulate)

→ **Storytelling**

1. **Descriptive Analytics**

- It is the examination of data or content, usually manually performed, to answer the question “**What happened?**” or “**What is happening?**”
- It is characterized by traditional BI and visualizations such as pie charts, bar charts, line graphs, tables, reporting, scorecards, clustering or generated narratives.

2. **Diagnostic Analytics**

- It examines data or content to answer the question
 “Why did it happen?”
- It is characterized by techniques such as drill-down, data discovery, data mining and correlations

3. **Predictive Analytics**

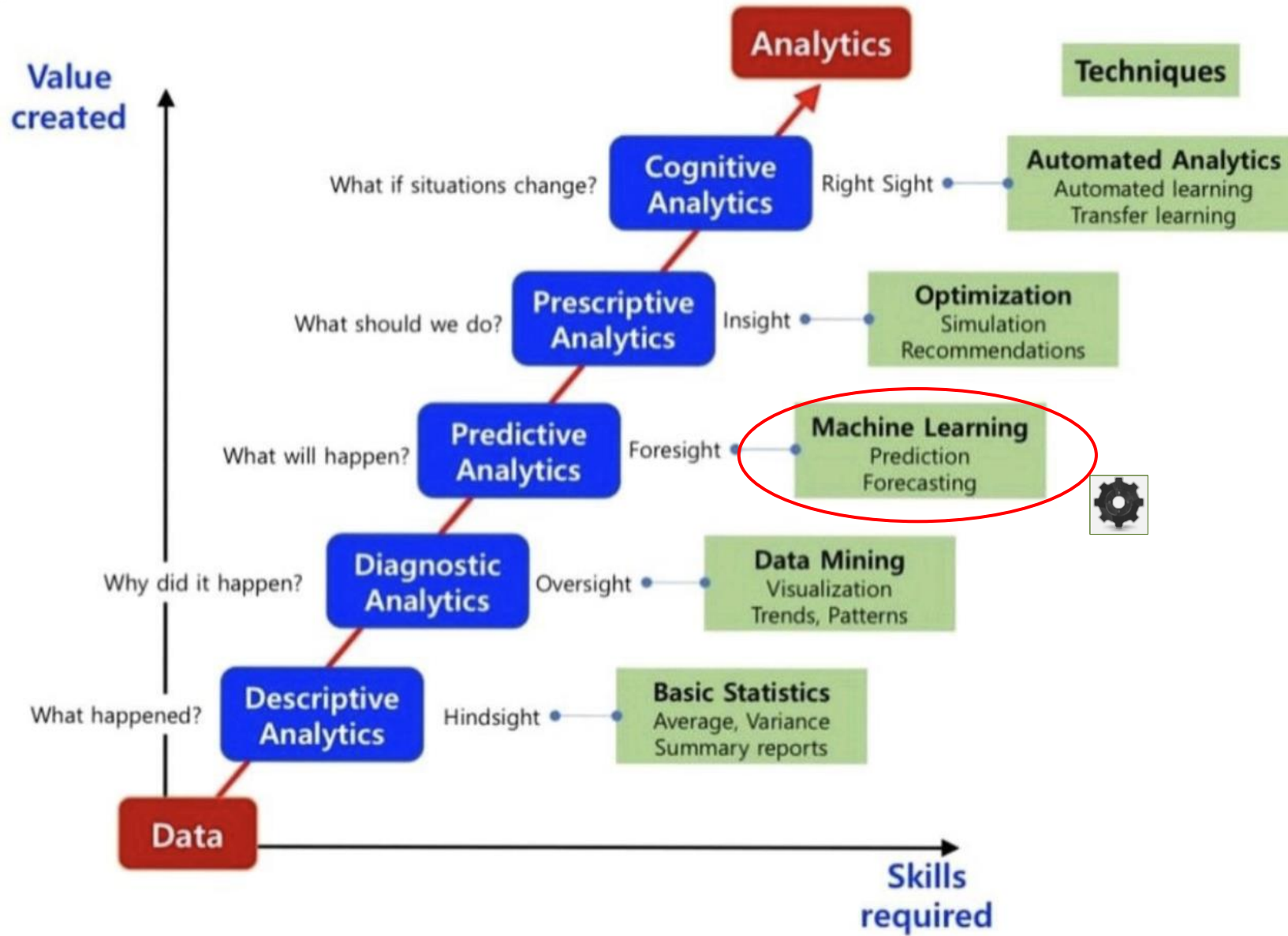
- It use statistical and **machine learning** techniques to answer the question “**What is likely to happen?**”.
- Predictive models exploit **trends and patterns** found in historical and transactional data to identify risks and opportunities and determine a **predictive score** (probability) for each associated entity

4. **Prescriptive Analytics**

- It examines data or content to answer the question “**What should be done?**” or “What can we do to make it happen?”
- It is characterized by techniques such as graph analysis, simulation, optimization, complex event processing, neural networks, heuristics, and machine learning.

5. Cognitive Analytics

- It examines data or content to answer the question “**What if situation changes?**”
- It is characterized by techniques such as automated analytics, automated learning and transfer learning
- Automated analytics are a form of advanced analytics that use emerging technology, such as machine learning, artificial intelligence, and natural language processing, to assist human data scientists and analysts with tedious tasks like data gathering and preparation.



Analytics – Human input vs Process automation

